Markets Need Memory

Human Contribution, Artificial Intelligence, and the Institutional Conditions of Legitimate Markets

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Abstract

Modern markets are increasingly asked to govern value created through long chains of human judgment, institutional support, public knowledge, data, software, and machine inference. Yet the institutions that recognize ownership and compensation usually preserve the final rights holder more reliably than the production history from which the asset emerged. Artificial intelligence intensifies this mismatch. It can aggregate, transform, and redeploy intermediate human contributions at a scale that makes those contributions economically important while making their individual histories harder to reconstruct after the fact.

This paper advances a public-interest claim rather than a technical claim about any single PBS/MPS implementation. Legitimate markets need institutions of economic memory: governed systems capable of preserving relevant contribution evidence before society decides whether recognition, allocation, ownership, licensing, compensation, restitution, or no further consequence should follow. Provenance does not itself establish desert or title. It preserves the factual conditions under which competing moral, legal, contractual, and economic accounts can deliberate without avoidable institutional amnesia.

The argument is conditional. More recording is not always better. Provenance can impose administrative cost, invade privacy, enable surveillance, harden error, encourage metric gaming, fragment claims, and commodify activities that should remain outside markets. The institutional response should therefore begin with comparison, adversarial testing, bounded pilots, independent replication, purpose limitation, correction rights, and domain-specific refusal criteria. The conclusion is nevertheless firm: as economic production becomes more cumulative, intangible, and machine-mediated, failing even to investigate credible forms of contribution memory creates risks for market legitimacy, bargaining fairness, innovation policy, dispute resolution, and public trust.

I. The Structural Mismatch

Societies increasingly confront a mismatch between how value is created and how economic institutions remember it. Value is often created cumulatively by people, data, institutions, inherited knowledge, public infrastructure, organizational routines, and machines. Ownership and compensation systems, by contrast, are designed to identify the final legal holder, contracting party, employer, assignee, or platform. That compression is necessary for transfer and settlement, but it can obscure the productive history that later disputes require.

> **The public problem is not that markets recognize final rights. It is that final rights can become the only surviving account of how value was created.**

A functioning market needs clear title, enforceable contracts, settled payment, and legal finality. This paper does not challenge those functions. It asks a prior governance question: when production is cumulative and the intermediate history may later matter, what institution preserves enough evidence for society to evaluate downstream claims responsibly?

1. Cumulative creation

A clinical model may depend on annotated cases, diagnostic protocols, hospital infrastructure, software libraries, prior research, regulatory standards, and later engineering. A seed variety may depend on inherited germplasm, public breeding programs, farmer selection, field trials, and commercial distribution. A creative model may depend on works, genres, technical tools, cultural conventions, and human

feedback. In each case, the terminal product is real, but it is not a complete description of the production process.

2. Compressed ownership records

Property and contract institutions intentionally compress history. A share records an ownership interest, not every act that produced enterprise value. An employment agreement allocates work-product rights without reconstructing every causal contribution. A platform license can authorize use without preserving the full dependency chain. This compression lowers transaction costs, but the omitted information can later become relevant to attribution, bargaining, licensing, liability, public funding, or rectification.

II. Why Artificial Intelligence Raises the Stakes

Artificial intelligence does not create the underlying problem, but it changes its scale and timing. Machine-learning systems can absorb large bodies of human-produced data, judgments, corrections, patterns, methods, and examples; transform them into model behavior; and deploy the resulting capability far from the original context. Once the source material has been aggregated, filtered, weighted, and encoded, reconstructing the contribution history may become technically difficult, institutionally costly, or impossible.

1. Visibility can decline as productive dependence increases

AI systems may make an upstream contribution more economically consequential while making the contributor less visible. The model is presented as the operative asset. The organization that owns or deploys it becomes the recognized actor. The earlier annotations, corrections, heuristics, examples, and institutional supports can disappear into a training pipeline or model update. This is not proof that every source deserves payment. It is a reason to preserve evidence before the question becomes unanswerable.

2. Machine mediation complicates causal narratives

A downstream output may depend on a contribution without reproducing it, and model performance may depend on combinations of inputs rather than a single source. Some contributions are redundant; others are complementary; some matter only under a particular architecture or baseline. These conditions make simplistic creator-to-output stories unreliable. They also make the absence of evidence more consequential, because later institutions cannot distinguish material dependence from rhetorical assertion without a record.

3. Scale magnifies governance error

A mistaken attribution rule applied once is a local error. A mistaken rule embedded in a large platform, procurement system, labor market, or public program can distribute error repeatedly. The same is true of omission. If contribution evidence is never preserved, systematic under-recognition can become administratively invisible. The public-interest case is therefore for governable evidence, not automatic compensation.

III. The Five-Step Public-Interest Argument

1. Value is created cumulatively.

Modern production combines human agency, data, institutions, knowledge, capital, software, and machine processes. The relevant unit of creation is often a dependency network rather than an isolated actor.

2. Rights systems identify terminal holders.

Legal and market institutions need a recognized holder for transfer, enforcement, settlement, and accountability. They are not generally designed to preserve the complete causal history of productive value.

3. AI and intangible production obscure intermediate contributions.

Reusable knowledge and machine-mediated production separate the time and place of contribution from later deployment, revenue, or social effect.

4. Fair attribution cannot be debated after the evidence disappears.

Moral, legal, contractual, and economic theories may disagree about consequence, but they require facts about participation, dependence, consent, transfer, performance, and use.

5. Society therefore needs institutions of economic memory.

The institutional task is to preserve relevant and contestable evidence before deciding whether rights, compensation, recognition, regulation, or no additional consequence should follow.

IV. Institutional Memory and Legitimate Markets

Market legitimacy does not require agreement that every contributor deserves an economic share. It requires that institutions do not present avoidable ignorance as neutrality. A market can recognize existing title while also preserving evidence relevant to whether that title, contract, allocation, or bargaining process should be examined under the applicable law or governance rule.

1. Legitimacy requires distinguishable stages

The canonical sequence remains: Provenance -> Performance -> Attribution/Allocation -> Entitlement/Payment. Provenance records relevant history. Performance supplies observable outcome evidence. Attribution estimates how documented inputs relate to outcomes. Allocation applies an explicit rule. Entitlement requires a legal, contractual, moral, regulatory, or governance warrant. Payment executes that warrant. No stage automatically becomes the next.

2. Shared evidence without shared doctrine

Institutional memory does not settle the dispute between desert, liberty, equality, need, efficiency, consent, and historical entitlement. It creates a disciplined factual ground on which those disagreements can occur. A property-rights account may use the evidence to examine acquisition and transfer. A labor or desert account may examine agency and contribution. A public-investment account may examine shared infrastructure. A court may conclude that no remedy follows. The record remains useful because the disagreement is no longer forced to rest on a missing history.

3. Markets need memory before they need a verdict

The immediate institutional objective should not be universal tokenization, a compulsory royalty, or a score of human worth. It should be the preservation of evidence whose later loss would predictably impair legitimate decision-making. Economic memory is infrastructure precisely because it can support multiple lawful outcomes, including the decision that no additional right or payment is justified.

V. Societal Risks of Institutional Amnesia

1. Bargaining power follows control of the record

When only one party controls the surviving records of creation and use, that party possesses an informational advantage in negotiation, employment mobility, licensing, and dispute resolution. Portable, contestable evidence can improve the outside option without guaranteeing a favorable outcome to the contributor. It can also disprove exaggerated claims.

2. Innovation policy becomes less informed

Governments, firms, universities, and funders cannot design sound incentives if they cannot identify which upstream inputs produce persistent downstream value. The risk is not only under-reward. Over-rewarding visible terminal actors can also misdirect investment and encourage strategic capture of credit.

3. Disputes harden into narratives

Where evidence is absent, disputes about creation can become contests of institutional power, reputation, or rhetoric. Provenance cannot eliminate conflict, but it can narrow factual disagreement, disclose uncertainty, and create a correction path.

4. Public trust deteriorates

A market order becomes harder to defend when people experience value creation as collective and cumulative but observe recognition and reward attached only to final possession. That perception does not prove injustice in every case. It does create a legitimacy risk when institutions cannot explain, with evidence, how the recognized allocation relates to the underlying history.

5. AI governance may focus on outputs while ignoring inputs

Safety, competition, privacy, and accountability remain essential AI-governance concerns. A complete framework should also ask whose documented contributions entered the system, under what terms, through which dependencies, with what consent or authority, and with what later use. Provenance adds an evidentiary dimension without predetermining the legal answer.

VI. What an Institution of Economic Memory Must Do

A legitimate provenance institution should be evaluated by function rather than by a preferred technology. Blockchain, conventional databases, registries, attestations, audit logs, contractual records, and privacy-preserving proofs are implementation options. The institutional requirements are more basic.

Requirement	Institutional meaning
Relevance	Preserve information material to a defined decision, not every observable fact.
Verifiability	Record source, date, method, status, and limitations.
Contestability	Provide notice, access, correction, response, appeal, and adjudication.

Uncertainty	Separate assertion, observation, inference, estimate, dispute, and determination.
Purpose limitation	Prevent attribution records from silently becoming surveillance, credit scoring, employment exclusion, or compulsory human ranking.
Privacy and security	Use selective disclosure, minimization, retention limits, and proportionate access.
Reversibility	Preserve audit history without making error institutionally immutable.
Independence	Protect measurement, validation, and adjudication from control by interested parties.
Proportionality	Match the recording burden and consequence to the value and reliability of the evidence.
Domain validity	Use methods appropriate to the epistemic structure of the field.

VII. What Institutional Memory Must Not Become

The strongest public-interest case for provenance includes refusal criteria. The architecture should be constrained or rejected when it creates a greater threat to dignity, autonomy, equality, experimentation, collaboration, or security than the informational problem it addresses.

- A universal score of personal worth.
- A compulsory identity system required for housing, employment, benefits, citizenship, or equal legal standing.
- A presumption that documentation creates ownership or payment.
- A system that records intimate behavior because collection is technically possible.
- An immutable repository of uncorrectable allegations or model outputs.
- A device for converting every social contribution into a market commodity.
- A pretext for replacing courts, regulators, collective bargaining, or democratic judgment with automated allocation.
- A requirement to measure domains in which valid baselines or causal models do not exist.

VIII. PBS/MPS as a Hypothesis, Not a Foregone Conclusion

PBS/MPS is one proposed architecture for supplying provenance, performance evidence, contribution decomposition, and later economic participation. The public-interest argument in this paper does not depend on the success of every PBS instrument or on the use of tokenization. The larger claim is institutional: societies should investigate whether governed contribution memory improves legitimate decision-making in domains where existing records are inadequate.

This separation also protects the framework from overclaiming. A ledger is not ownership. A performance measure is not attribution. An attribution model is not exact causal truth. A Shapley allocation is conditional on the specified value function and adopted axioms. A royalty is not created by mathematics. The Exchange Protocol, where used, is surrounding market infrastructure rather than a substitute for the analytical sequence.

IX. A Humble Research and Governance Program

1. Comparative institutional analysis

Proposed systems should be compared with existing contracts, registries, professional records, collective licensing, audit systems, courts, arbitration, and organizational governance. The relevant question is not whether a new provenance system is more complete. It is whether it improves decisions, fairness, cost, speed, contestability, or trust relative to realistic alternatives.

2. Adversarial testing

Testing should assume strategic behavior. Contributors may fragment claims, inflate importance, coordinate attestations, optimize metrics, conceal dependencies, or challenge unfavorable records. Asset holders may suppress use data, define narrow baselines, exploit opacity, or control validation. Red-team evaluation should examine both directions of manipulation.

3. Bounded pilots

Pilots should begin in domains with identifiable contributions, observable outcomes, meaningful materiality thresholds, manageable privacy risk, and institutions capable of correction and appeal. A pilot should specify in advance what consequence, if any, the evidence may support. Recording should not silently expand into entitlement.

4. Independent replication

Claims about attribution accuracy, transaction-cost reduction, labor mobility, licensing, innovation, or dispute resolution should be replicated by parties independent of the system designer and economically interested participants. A framework that cannot survive independent evaluation should not govern consequential rights.

5. Explicit stopping rules

A pilot should stop or narrow when measurement error remains too high, participants cannot contest records, administrative cost exceeds informational benefit, privacy risk cannot be contained, strategic behavior overwhelms the metric, or the proposed consequence exceeds the evidence.

X. A Testable Public-Interest Agenda

- Do contribution records improve the quality and speed of licensing, employment, investment, or dispute decisions?
- Do portable records reduce dependence on the institution that controls the original files?
- Which forms of contribution history remain useful without creating a property right?
- When does preservation improve innovation incentives, and when does it create claim fragmentation?
- Can independent reviewers reproduce attribution results from the disclosed evidence and assumptions?
- How do privacy-preserving designs compare with centralized or public ledgers?

- What materiality threshold keeps administrative cost below social benefit?
- Does provenance increase trust only when meaningful correction and appeal rights exist?
- Which domains should remain formally unmeasured?

XI. Objections and Replies

"Markets need clear ownership, not endless history."

They need both clear ownership and rules about when history is relevant. Economic memory should be bounded by materiality, legal finality, proportionality, and cost. Its purpose is not to reopen every transaction. It is to prevent consequential questions from becoming impossible merely because the evidence was never preserved.

"Provenance will produce endless claims."

A record is not a claim, and a claim is not an entitlement. Burdens of proof, limitation periods, consent, reliance, administrability, and remedial proportionality remain necessary. Better evidence can support rejection as well as recognition.

"Measurement will be biased or gamed."

That risk is real. It supports adversarial testing, multiple evidence streams, independent validation, uncertainty disclosure, correction rights, and authority to decline attribution. It does not support automatic blindness where more reliable evidence is feasible.

"The project commodifies people."

It would if contribution records became compulsory identities or measures of human worth. The framework rejects that use. Persons are not assets. Dignity, citizenship, rights, care, and social membership are not conditional on measured economic contribution.

"Existing law already allocates the rights."

Existing law may allocate rights conclusively. Provenance does not override that allocation. It can inform future contracting, policy, licensing, evidence, or reform while preserving the distinction between recognized title and productive history.

XII. The Unequivocal Claim

Legitimate markets require more than memory of possession. Where cumulative and machine-mediated production makes contribution history materially relevant, society should develop governed means of preserving that history before deciding what rights or compensation, if any, should follow.

This claim is deliberately narrower than a universal right to ownership, royalty, or tokenization. It requires investigation, not predetermined adoption. It requires restraint, not maximal recording. But it rejects one position as institutionally inadequate: that the disappearance of contribution evidence is neutral merely because final title remains clear.

Conclusion

The infrastructure of a market is not limited to exchanges, payment rails, contracts, registries, and courts. It also includes the records through which institutions can explain how recognized assets, capabilities, and claims came to exist. In a production system shaped by cumulative innovation, distributed expertise, intangible assets, and artificial intelligence, that explanatory capacity is becoming economically and politically consequential.

Society does not need a ledger that decides justice. It needs institutions capable of remembering enough for justice, law, contract, and policy to decide without avoidable amnesia. Provenance should therefore be developed as public-interest evidence infrastructure: selective, contestable, privacy-preserving, corrigible, independently tested, and strictly separated from automatic entitlement.

The failure to build such institutions may leave markets efficient at transfer but increasingly unable to justify their allocations to the people whose contributions made those allocations possible. The failure even to investigate them would accept institutional forgetting without first measuring its social cost.

Claim-Status Note

This paper is an author synthesis and normative public-interest proposal. It does not claim that PBS/MPS is a deployed or independently validated production system, that comprehensive provenance is desirable, that attribution is universally measurable, that contribution evidence creates ownership, or that any particular technology is necessary. The proposed research posture is comparative, adversarial, bounded, independently replicable, and open to refusal where costs or risks exceed likely benefit.

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